

The $\overline{\text{MS}}$ row in transverse position space

Trading $\log \frac{\mu^2}{k^2}$ for the dipole size, and what it costs

$$r = z - x, \quad s = y - x, \quad q = \bar{\xi}k, \quad D = (r - s)^2 + M^2, \quad M^2 = \frac{\xi}{\bar{\xi}}s^2, \quad D_0 = D|_{r=0} = \frac{s^2}{\bar{\xi}}$$

1. What is actually in momentum space

Only one thing: the argument of the logarithm. The row itself — the Wilson lines, the colour sources, the kernel $\frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2}$ — is already a transverse-position-space object. What came out of $\overline{\text{MS}}$ was

$$T|_{\overline{\text{MS}}} = \pi \log \frac{\mu^2}{q^2}, \quad q = \bar{\xi}k,$$

i.e. the produced-gluon momentum, not the dipole size.

This is not a rewriting. You cannot turn $\log \frac{\mu^2}{k^2}$ into $\log (\mu^2 (y-x)^2)$ by algebra — the two differ by $\log (k^2 (y-x)^2)$, which is a real function of the kinematics. The conversion needs the piece the draft you are holding left out: the finite remainder

$$\mathcal{R} = \int d^2r \, e^{iq \cdot r} \frac{1}{r^2} \left[\frac{1}{D} - \frac{1}{D_0} \right].$$

It is ultraviolet finite and needs no regulator, so it is easy to forget — but it carries a $\log q^2$ of its own, of the opposite sign, and that is what cancels the q^2 .

2. Regulate in position space to begin with

The cleanest route never leaves position space. Cut the transverse integral at short distance, $|r| > r_0$. Then

$$\int_{|r|>r_0} d^2r \, \frac{e^{iq \cdot r}}{r^2} = 2\pi \int_{r_0}^{\infty} \frac{dr}{r} J_0(qr) = \pi \log \frac{4e^{-2\gamma_E}}{q^2 r_0^2} + O(q^2 r_0^2).$$

The mass regulator $\frac{1}{r^2} \rightarrow \frac{1}{r^2 + m^2}$ gives $2\pi K_0(m|q|)$, which is *the same expression with $m = r_0$* . Checked:

$ q $	$r_0 = m$	disc cut (quadrature)	$2\pi K_0(m q)$	$\pi \log \frac{4e^{-2\gamma_E}}{q^2 r_0^2}$
0.60	10^{-3}	47.340738007	47.340742550	47.340737724
0.60	10^{-4}	61.808306552	61.808306610	61.808306549
1.40	10^{-3}	42.017009796	42.017031924	42.017008257
1.40	10^{-4}	56.484577097	56.484577389	56.484577082

The $\overline{\text{MS}}$ dictionary

$$\pi \log \frac{\mu^2}{q^2} = \pi \log \frac{4e^{-2\gamma_E}}{q^2 r_0^2} \quad \Longleftrightarrow \quad \boxed{r_0 = \frac{2e^{-\gamma_E}}{\mu}}$$

(μ the $\overline{\text{MS}}$ scale). Exact to all digits, both signs of the log — this is the only place γ_E and $\log 2$ ever enter.

3. The master, entirely in position space

$$G = \int d^2r \, \frac{e^{iq \cdot r}}{r^2 D} = \frac{\pi}{D_0} \left[\log \frac{D_0}{r_0^2} + \hat{I}(\xi) \right], \quad D_0 = \frac{(y-x)^2}{\xi},$$

$$\hat{I}(\xi) = D_0 \int_0^1 dt \left[e^{itq \cdot s} \frac{|q| K_1(|q| \sqrt{\Delta_0})}{\sqrt{\Delta_0}} - \frac{1}{t D_0} \right], \quad \Delta_0 = t[(1-t)s^2 + M^2].$$

No γ_E , no $\log 2$, no q^2 — the scale in the logarithm is the dipole size divided by $\bar{\xi}$, and the regulator sits where it belongs, as a transverse distance. $\hat{I}$ carries no regulator and no μ . Verified against direct two-dimensional quadrature of $\int_{|r|>r_0}$:

ξ	r_0	quadrature	formula	rel
0.30	10^{-2}	$33.327671989 + 1.388454714i$	$33.327769233 + 1.388554182i$	4.2×10^{-6}
0.30	10^{-3}	$48.908226995 + 1.388553187i$	$48.908227968 + 1.388554182i$	2.8×10^{-8}
0.65	10^{-2}	$16.917393275 + 0.257745919i$	$16.917359887 + 0.257758352i$	2.1×10^{-6}
0.65	10^{-3}	$24.707589587 + 0.257758228i$	$24.707589254 + 0.257758352i$	1.4×10^{-8}

Substituting $r_0 = 2e^{-\gamma_E}/\mu$ recovers the $\overline{\text{MS}}$ form,

$$\log \frac{D_0}{r_0^2} = \log \frac{\mu^2(y-x)^2}{\bar{\xi}} + 2(\gamma_E - \log 2),$$

exact to twelve digits. So the celebrated $2(\gamma_E - \log 2)$ is *nothing but the $\overline{\text{MS}}$ translation constant*: in position space with a distance cutoff it is not there at all.

4. The p^+ integral, and the answer

Split $\log \frac{D_0}{r_0^2} = \log \frac{(y-x)^2}{r_0^2} + \log \frac{1}{\xi}$; the first factor is ξ -independent. With $C_{\text{UV}}(\xi) = \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}}{2N_c}$ and $\ell_k = \log \frac{k^+}{\Lambda}$,

$$\begin{aligned} \int d\xi C_{\text{UV}} &= 2\ell_k - \frac{11}{6}, & \int d\xi C_{\text{UV}} \log \frac{1}{\xi} &= \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}, \\ \int d\xi C_{\text{UV}} \hat{I} &= \frac{e^{ik \cdot (y-x)}}{2} \ell_k^2 + c_0 \ell_k + c_1. \end{aligned}$$

The row in transverse position space. With $\frac{d^3 N_2^{\text{NLO}}}{d^3 k} = \frac{\alpha_s^2 N_c}{8\pi^5 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} [\mathcal{P} + \mathcal{F}] \times$
(Wilson lines),

$$\mathcal{F} = \left(2\ell_k - \frac{11}{6}\right) \log \frac{(y-x)^2}{r_0^2} + \frac{1 + e^{ik \cdot (y-x)}}{2} \ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36} + c_0 \ell_k + c_1,$$

and if you insist on $\overline{\text{MS}}$, put $r_0 = 2e^{-\gamma_E}/\mu$:

$$\mathcal{F} = \left(2\ell_k - \frac{11}{6}\right) \left[\log (\mu^2(y-x)^2) + 2(\gamma_E - \log 2) \right] + \frac{1 + e^{ik \cdot (y-x)}}{2} \ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36} + c_0 \ell_k + c_1.$$

Two readings. First, $(2\ell_k - \frac{11}{6}) \times 2N_c = 4N_c \ell_k - b$ at $N_f = 0$, so the constant riding on it enters as $2b(\gamma_E - \log 2)$ — exactly the term in K_{JSJ} , Eq. (2.64) of arXiv:1610.03453 that their footnote 2 records as having been missing. Second, the ℓ_k^2 coefficient is $\frac{1}{2}(1 + e^{ik \cdot (y-x)})$, one half from each endpoint of the ξ integral; it is regulator-independent and agrees with the mass-regulator computation.

On the draft you attached. It is the three-page version, which kept only T and therefore states that “the scale in the logarithm is k^2 , not the dipole size”. That reading is superseded: once $\mathcal{R}$ is kept the q^2 cancels and the logarithm is of the dipole size. The current four-page `dimreg_msbar.pdf` has §§6–9 rewritten accordingly.

`zint/src/posspace.py` — every number above, written to `zint/RESULTS_posspace.txt`; `zint/src/uv_master.py`, `zint/src/remainder.py` — $\hat{I}$ and $\mathcal{R}$.